Lab program 3.c - Code::Blocks 25.03

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```c
#include<stdio.h>
#define MAX5
int stack[MAX5];
int top=-1;
void push(int value){
if(top==MAX5-1)
    printf("stack Overflow\n");
else{
    top++;
    stack[top] = value;
    printf("pushed %d\n",value);

}
}
void pop(){
if(top==-1)
    printf("stack underflow/n");
else{
    printf("popped %d/n",stack[top]);
    top--;
}
}
void display(){
if(top==-1)
    printf("stack is empty\n");
else{
    printf("stack elements:\n");
    for(int i=top;i>=0;i--);
       printf("%d\n",stack[i]);
}

}
int main(){
```

C:\Users\Sevanthi-v\Downloads\Lab program 3.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 29, Col 32, Pos 512 | Insert | Read/Write | default

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```c
    printf("stack Overflow\n");
else{
    top++;
    stack[top] = value;
    printf("pushed %d\n",value);

}
}
void pop(){
if(top==-1)
    printf("stack underflow/n");
else{
    printf("popped %d/n",stack[top]);
    top--;
}
}
void display(){
if(top==-1)
    printf("stack is empty\n");
else{
    printf("stack elements:\n");
    for(int i=top;i>=0;i--);
       printf("%d\n",stack[i]);
}

}
int main(){
push(10);
push(20);
pop();
}
```

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lab program DS1.c - Code::Blocks 25.03

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```c

/*implement basic array operations:insertion,deletion,searching,and traversal*/
#include<stdio.h>
int main(){
int arr[100],n,i,pos,value,choice,found;
printf("Enter number of element:");
scanf("%d",&n);
printf("Enter element;\n");
for(i=0;i<n;i++){
    scanf("%d",&arr[i]);
}
do{
    printf("\n---MENU---\n");
    printf("1.Traversal\n2.Insertion\n3.Deletion\n4.searching\n5.Exit\n");
    printf("Enter your choice");
    scanf("%d",&choice);
    case1:
        printf("Array elements are:");
        for(i=0;i<n;i++){
            printf("%d",arr[i]);
}
   break;
    case2:
        printf("Enter positive and value:");
        scanf("%d %d",&pos,&value);
        for(i=0;i>=pos;i--){
            arr[i]=arr[i-1];
        }
        arr[pos-1]=value;
        n++;
        break;
   case3:
   printf("Enter position to delete:");
   scanf("%d",pos);
   for(i=pos-1;i<n-1;i++){
```

C:\Users\Sevanthi-v\Downloads\lab program DS1.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 56, Col 19, Pos 1376 | Insert | Read/Write | default

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Start here | Lab program 3.c | lab program DS1.c | lab program DS2.c | DS lab 9.c

```
        n++;
        break;
   case3:
   printf("Enter position to delete:");
   scanf("%d",pos);
   for(i=pos-1;i<n-1;i++){
    arr[i]==arr[i+1];
   }
   n--;
   break;
   case4:
       printf("Enter element to search");
       scanf("%d",value);
       found=0;
       for(i=0;i<n;i++){
        if(arr[i]== value){
            printf("element found position:",i+i);
            found=1;
            break;
        }
       }
       if(found==0)
        printf("element not found/n");
        break;
        case5:
            printf("Exiting.../n");
            break;
            defult:
            printf("invalid choice!/n");
}
while(choice !=5);
return 0;
}
```

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lab program DS2.c - Code::Blocks 25.03

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Start here | Lab program 3.c | lab program DS1.c | lab program DS2.c | DS lab 9.c

```c

#include<stdio.h>
int main(){
int matrix[10][10],transpose[50][3],triplet[50][3];
int i,j,m,n,k=1;

printf(" enter rows and coloums:");
scanf("%d %d ",&m,&n);
printf(" enter matrix elements");
for(i=0;i<m;i++){
    for(j=0;j<n;j++){
            scanf("%d",&matrix[i][j]);

    }
}

triplet[0][0]=m;
triplet[0][1]=n;
triplet[0][2]=0;
for(i=0;i<m;i++){
    for(j=0;j<n;j++){
        if(matrix[i][j]!=0){
            triplet[k][0]=i;
            triplet[k][1]=j;
            triplet [k][2]=matrix[i][j];
            k++;
            triplet[0][2]++;
        }
    }
}
printf("\nTriple Representation:");
for(i=0;i<k;i++){
    printf("%d %d %d\n",triplet[i][0],triplet[i][1],triplet[i][2]);

}
```

C:\Users\Sevanthi-v\Downloads\lab program DS2.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 23, Col 29, Pos 453 | Insert | Read/Write | default

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```c
            triplet[k][1]=j;
            triplet [k][2]=matrix[i][j];
            k++;
            triplet[0][2]++;
        }
    }
}
printf("\nTriple Representation:");
for(i=0;i<k;i++){
    printf("%d %d %d\n",triplet[i][0],triplet[i][1],triplet[i][2]);

}

transpose[0][0]=triplet[0][1];
transpose[0][1]=triplet[0][0];
transpose[0][2]=triplet[0][2];
k=1;
for (i=0;i<n;i++){
    for(j=1;j<triplet[0][2];j++){
        if(triplet[j][1]==i){
            transpose[k][0]=triplet[j][1];
            transpose[k][1]=triplet[j][0];
            transpose[k][2]=triplet[j][2];
            k++;
        }
    }
}
printf("\nTranspose triplet/n");
for(i=0;i<k;i++){
    printf("%d %D %d", transpose[i][0],transpose[i][1],transpose[i][2]);
}
return 0;
}
```

C:\Users\Sevanthi-v\Downloads\lab program DS2.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 23, Col 29, Pos 453 | Insert | Read/Write | default

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Start here | Lab program 3.c | lab program DS1.c | *lab program DS2.c | DS lab 9.c

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* next;
};


struct Node* createNode(int data) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        exit(1);
    }

    newNode->data = data;
    newNode->next = NULL;
    return newNode;
}


void insertAtBeginning(struct Node** head, int data) {
    struct Node* newNode = createNode(data);
    newNode->next = *head;
    *head = newNode;
}


void insertAtEnd(struct Node** head, int data) {
    struct Node* newNode = createNode(data);

    if (*head == NULL) {
        *head = newNode;
```

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Start here | Lab program 3.c | lab program DS1.c | lab program DS2.c | DS lab 9.c

```c
    if (*head == NULL) {
        *head = newNode;
        return;
    }

    struct Node* temp = *head;
    while (temp->next != NULL) {
        temp = temp->next;
    }

    temp->next = newNode;
}


void insertAtPosition(struct Node** head, int data, int position) {
    if (position < 0) {
        printf("Invalid position\n");
        return;
    }

    if (position == 0) {
        insertAtBeginning(head, data);
        return;
    }

    struct Node* newNode = createNode(data);
    struct Node* temp = *head;

    for (int i = 0; temp != NULL && i < position - 1; i++) {
        temp = temp->next;
    }

    if (temp == NULL) {
        printf("Position out of range\n");
        free(newNode);
```

C:\Users\Sevanthi-v\Downloads\DS lab 9.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 62, Col 1, Pos 1202 | Insert | Read/Write | default

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Start here | Lab program 3.c | lab program DS1.c | lab program DS2.c | DS lab 9.c

```
        printf("Position out of range\n");
        free(newNode);
        return;
    }

    newNode->next = temp->next;
    temp->next = newNode;
}


void deleteFromBeginning(struct Node** head) {
    if (*head == NULL) return;


    struct Node* temp = *head;
    *head = (*head)->next;
    free(temp);
}

void deleteFromEnd(struct Node** head) {
    if (*head == NULL) return;

    if ((*head)->next == NULL) {
        free(*head);
        *head = NULL;
        return;
    }

    struct Node* temp = *head;

    while (temp->next->next != NULL) {
        temp = temp->next;
    }

    free(temp->next);
```

C:\Users\Sevanthi-v\Downloads\DS lab 9.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 62, Col 1, Pos 1202 | Insert | Read/Write | default

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```c

    free(temp->next);
    temp->next = NULL;
}


void deleteFromPosition(struct Node** head, int position) {
    if (*head == NULL || position < 0) return;

    if (position == 0) {
        deleteFromBeginning(head);
        return;
    }

    struct Node* temp = *head;

    for (int i = 0; temp != NULL && i < position - 1; i++) {
        temp = temp->next;
    }

    if (temp == NULL || temp->next == NULL) {
        printf("Position out of range\n");
        return;
    }

    struct Node* nodeToDelete = temp->next;
    temp->next = nodeToDelete->next;
    free(nodeToDelete);
}


void traverse(struct Node* head) {
    struct Node* temp = head;

    while (temp != NULL) {
```

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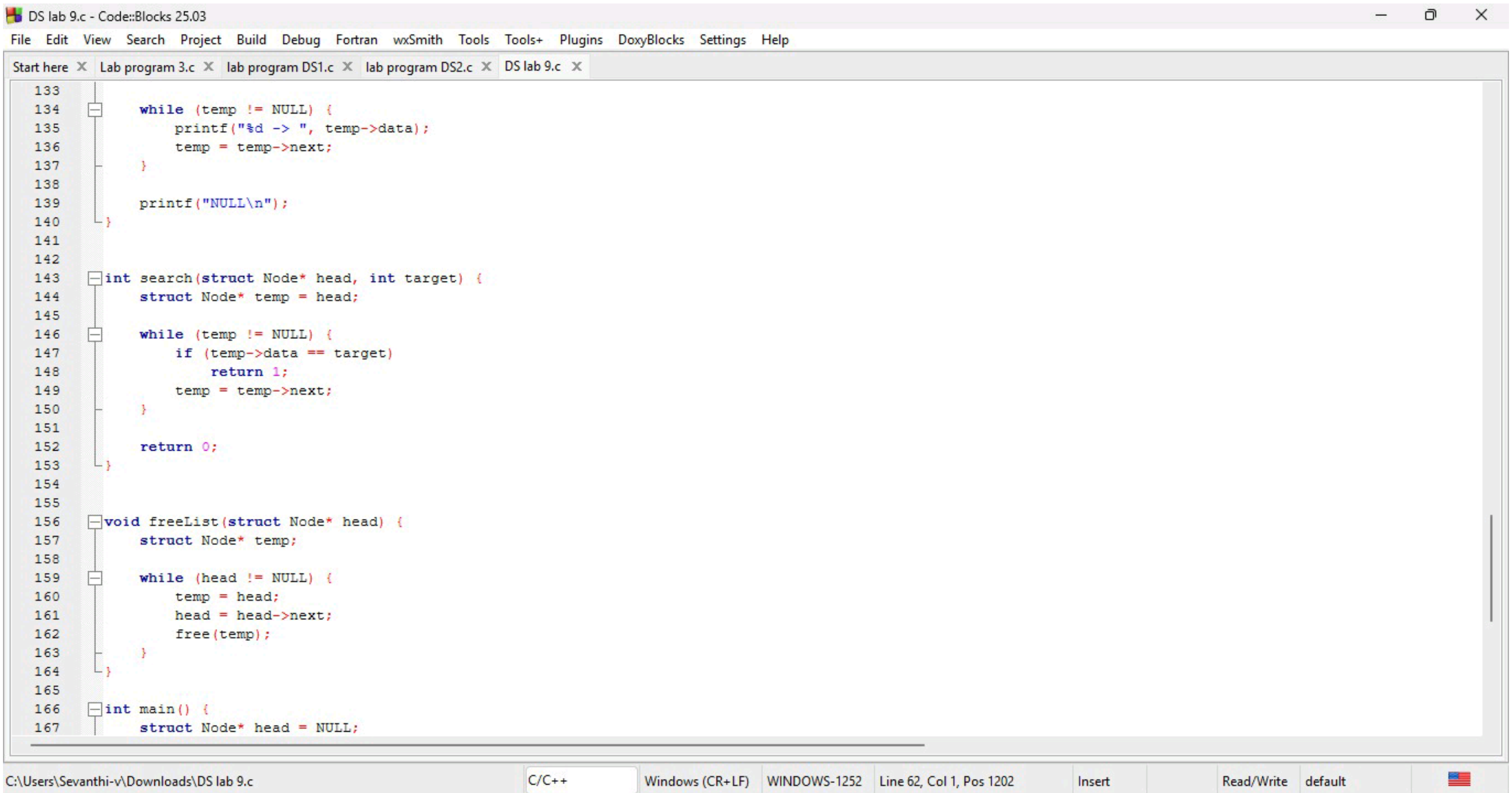
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Start here
Lab program 3.c
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DS lab 9.c
    while (temp != NULL) {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }

    printf("NULL\n");
}

int search(struct Node* head, int target) {
    struct Node* temp = head;

    while (temp != NULL) {
        if (temp->data == target)
            return 1;
        temp = temp->next;
    }

    return 0;
}

void freeList(struct Node* head) {
    struct Node* temp;

    while (head != NULL) {
        temp = head;
        head = head->next;
        free(temp);
    }
}

int main() {
    struct Node* head = NULL;
C:\Users\Sevanthi-v\Downloads\DS lab 9.c
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 62, Col 1, Pos 1202
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Read/Write
default

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Start here | Lab program 3.c | lab program DS1.c | lab program DS2.c | DS lab 9.c

```c
    }
}

int main() {
    struct Node* head = NULL;

    insertAtEnd(&head, 10);
    insertAtEnd(&head, 20);
    insertAtEnd(&head, 30);
    traverse(head);

    insertAtBeginning(&head, 5);
    traverse(head);

    insertAtPosition(&head, 25, 2);
    traverse(head);

    deleteFromBeginning(&head);
    traverse(head);

    deleteFromEnd(&head);
    traverse(head);

    deleteFromPosition(&head, 1);
    traverse(head);

    printf("List contains 20: %s\n", search(head, 20) ? "Yes" : "No");
    printf("List contains 40: %s\n", search(head, 40) ? "Yes" : "No");

    freeList(head);

    return 0;
}
```

C:\Users\Sevanthi-v\Downloads\DS lab 9.c | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 62, Col 1, Pos 1202 | Insert | Read/Write | default